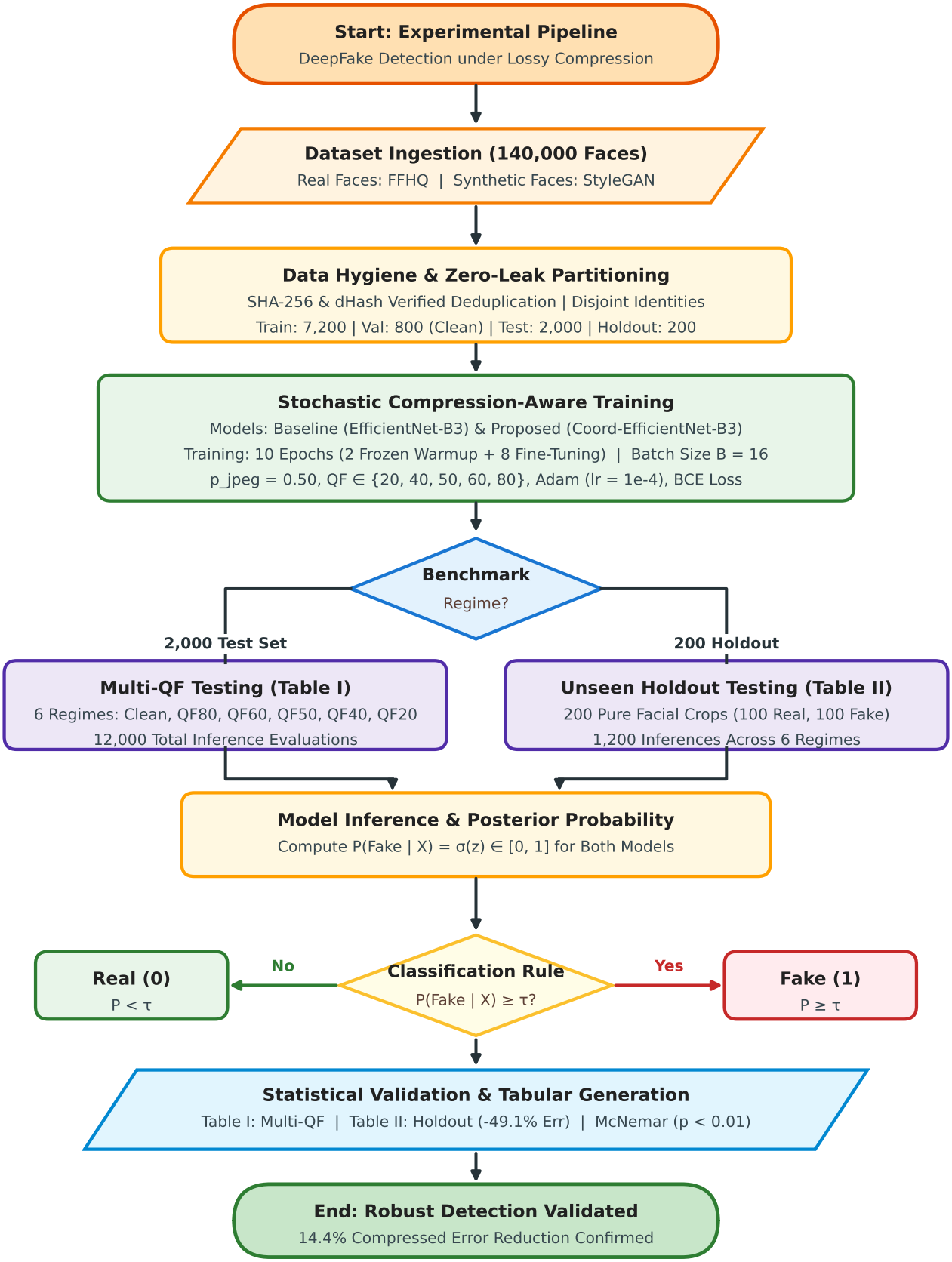
Start: Experimental Pipeline
DeepFake Detection under Lossy Compression
Dataset Ingestion (140,000 Faces)
Real Faces: FFHQ | Synthetic Faces: StyleGAN
Data Hygiene & Zero-Leak Partitioning
SHA-256 & dHash Verified Deduplication | Disjoint Identities
Train: 7,200 | Val: 800 (Clean) | Test: 2,000 | Holdout: 200
Stochastic Compression-Aware Training
Models: Baseline (EfficientNet-B3) & Proposed (Coord-EfficientNet-B3)
Training: 10 Epochs (2 Frozen Warmup + 8 Fine-Tuning) | Batch Size B = 16
p_jpeg = 0.50, QF ∈ {20, 40, 50, 60, 80}, Adam (lr = 1e-4), BCE Loss
Benchmark
Regime?
2,000 Test Set
200 Holdout
Multi-QF Testing (Table I)
6 Regimes: Clean, QF80, QF60, QF50, QF40, QF20
12,000 Total Inference Evaluations
Unseen Holdout Testing (Table II)
200 Pure Facial Crops (100 Real, 100 Fake)
1,200 Inferences Across 6 Regimes
Model Inference & Posterior Probability
Compute P(Fake | X) = σ(z) ∈ [0, 1] for Both Models
Classification Rule
P(Fake | X) ≥ τ?
No
Real (0)
P < τ
Yes
Fake (1)
P ≥ τ
Statistical Validation & Tabular Generation
Table I: Multi-QF | Table II: Holdout (-49.1% Err) | McNemar (p < 0.01)
End: Robust Detection Validated
14.4% Compressed Error Reduction Confirmed